

How to treat cancer in adults?

Evidence searched on 2026-10-09 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence does not provide direct clinical guidance on treating adult cancer, as no published studies or efficacy results were found. The only data available are **three ongoing or planned trials** (e.g., NCT07578649, a Phase 3 study of PF-08634404 + pembrolizumab + chemotherapy for advanced endometrial cancer, *not yet recruiting*) and **FDA approval announcements** (e.g., zenocutuzumab-zbco for NRG1-fusion NSCLC, *FDA, 2026 URL*), which do not establish broad treatment approaches.

Most important recent finding FDA granted accelerated approval to **zenocutuzumab-zbco (Bizengri®)** for adults with advanced, unresectable, or metastatic NSCLC harboring a *NRG1* gene fusion (*FDA, 2026*) [<https://www.lungcancerresearchfoundation.org/research/why-research/treatment-advances>].

Immediate context No direct evidence from published studies or completed trials was retrieved to support general cancer treatment strategies in adults. Approvals (e.g., tarlatamab for small cell lung cancer, *FDA, 2025*) are limited to specific biomarkers or cancer types and do not apply broadly. Ongoing trials explore targeted combinations (e.g., immunotherapy + chemotherapy), but results are pending.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Ondansetron	"Ondansetron is indicated for the prevention of nausea and vomiting associated with: highly emetogenic cancer chemotherapy, including cisplatin greater than or equal to 50 mg/m ² . Initial and repeat courses of ..."	DailyMed
PF-08634404 (latest, investigational)	Being tested in a registered Phase 3 trial, Not yet recruiting; not an approved treatment	NCT07578649

Latest findings

- **2027 Feb 1:** Wang *et al.* (GC-MS-based exhaled VOC diagnostic model for distinguishing early-stage lung cancer from controls) – A single-center study identified early metabolic alterations in exhaled breath that distinguish early-stage lung cancer from controls, but requires external validation for generalizability [3]. (*indirect evidence: different or related population*)
- **2027 Feb 1:** Li *et al.* (Peptide-functionalized magnetic nanobeads for EpCAM-targeted CTC isolation and immunotherapy efficacy evaluation in liver cancer) – A peptide-nanobead liquid biopsy strategy effectively stratifies liver cancer patients likely to benefit from PD-1 inhibitors, with cTPShigh as an independent predictor of progression-free and overall survival [1]. (*indirect evidence: different or related population*)
- **2027 Jan 15:** Odrzywolski *et al.* (Extracellular vesicle-derived FTIR spectroscopy for colorectal cancer detection) – FTIR spectroscopy of extracellular vesicles (EVs) achieved 95.6–97.8% accuracy in distinguishing colorectal cancer patients from controls, outperforming plasma-based methods [4]. (*indirect evidence: different or related population*)
- **Advance publication (issue dated 2027 Jan):** Allen *et al.* (Hypofractionated proton radiotherapy for high-risk lung cancer) – Hypofractionated proton radiotherapy demonstrated 1- and 2-year local control rates of 83.1% and 77.4%, respectively, in a high-risk lung cancer cohort [2].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Systemic immunotherapy (PD-1 inhibitors)	Indirect evidence (different or related population)	Different or related population (see the cited study); not condition-specific	[1]
Radiotherapy (hypofractionated proton therapy)	Experimental or preliminary	Hypofractionated proton radiotherapy achieved 83.1% 1-year and 77.4% 2-year local control in high-risk lung cancer patients, with no significant differences by tumor size, ultracentral location, or re-irradiation (2027).	[2]
Breath-based diagnostics (VOCs)	Indirect evidence (different or related population)	Different or related population (see the cited study); not condition-specific	[3]
Extracellular vesicle (EV) spectroscopy	Indirect evidence (different or related population)	Different or related population (see the cited study); not condition-specific	[4]
Music therapy for symptom management	Evidence-supported but limited	A systematic review of 6 controlled trials (n=438) found music therapy significantly reduced anxiety and improved spiritual well-being and ego-integrity in advanced cancer patients (2026).	[5]
Exercise interventions (geriatric oncology)	Emerging	Exercise remodels the aging microenvironment, improving prognosis in older cancer patients, though evidence is observational and lacks randomized trial data (2026).	[6]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Ondansetron	"Ondansetron is indicated for the prevention of nausea and vomiting associated with: • highly emetogenic cancer chemotherapy, including cisplatin greater than or equal to 50 mg/m ² • initial and repeat courses of moderately emetogenic cancer chemotherapy • radiotherapy in patients receiving either total body irradiation, single high-dose fraction to the abdomen, or daily fractions to the abdomen" [associated symptom or comorbidity, not the condition itself]	Route: Oral. No dosing section for this condition was found in the retrieved label text; see the linked label.	FDA-approved for the use shown; label effective 2026-07-29; original approval date not retrieved	FDA label. DailyMed label

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
tarlatamab	advanced small cell lung cancer after platinum-based chemotherapy (population: adults)	"Lung Cancer For adults with advanced small cell lung cancer that has spread after platinum-based chemotherapy, the FDA granted full approval on November 19 to tarlatamab (Imdelltra®), a targeted immunotherapy drug." MSK-Led Research Resulted in New FDA Approvals for Cancer Drugs in 2025 / Memorial Sloan Kettering Cancer Center
nirogacestat	progressing desmoid tumors in adults (population: adults)	"First, nirogacestat-a gamma secretase inhibitor-was approved for the treatment of adults with progressing desmoid tumors." A Comprehensive Overview of US-FDA-Approved Anticancer Drugs and Biologics in 2023: An Observational Study
talquetamab-tgvs	as stated in the quoted source sentence	"Second, talquetamab-tgvs, a bispecific GPRC5D-directed CD3 T-cell engager, was approved for the treatment of relapsed or ." A Comprehensive Overview of US-FDA-Approved Anticancer Drugs and Biologics in 2023: An Observational Study
toripalimab-tpzi	first-line metastatic or recurrent nasopharyngeal carcinoma (population: adults)	"FDA approves toripalimab-tpzi for nasopharyngeal carcinoma / On October 27, 2023, the Food and Drug Administration approved toripalimab-tpzi (LOQTORZ, Coherus BioSciences, Inc.) with cisplatin and gemcitabine for the first-line treatment of adults with metastatic or recurrent, locally advanced nasopharyngeal carcinoma (NPC)." Oncology (Cancer)/Hematologic Malignancies Approval Notifications / FDA

Recently approved or updated (FDA, EMA and CDSCO announcements)

Date	Type	Announcement	What the regulator's page says
September 04, 2026	Approval	FDA Grants Accelerated Approval to a New Breast Cancer Treatment / FDA	"Food and Drug Administration today expanded treatment options for adult patients with advanced breast cancer, reflecting the FDA's commitment to advancing medical innovation and getting new treatments to patients who need them.. "This marks the first FDA approval of a cancer therapy guided by the detection of a resistance mutation in circulating tumor DNA (ctDNA) before imaging tests show that the disease is progressing."
August 26, 2026	Approval	FDA Approves First in Class Targeted Therapy for Metastatic Pancreatic Cancer / FDA	"Food and Drug Administration today approved Rasonque (daraxonrasib), a RAS inhibitor for the most common form of pancreatic cancer-delivering a new treatment option to patients with advanced pancreatic cancer months ahead of schedule.. Rasonque, a tablet taken once daily, targets multiple forms of a protein called RAS, a key driver of tumor growth in most patients with pancreatic adenocarcinoma, which arises from cells lining the ducts of the pancreas.."
August 06, 2026	Approval	FDA Approves New Engineered Viral Immunotherapy for Patients with Treatment-Resistant Advanced Melanoma / FDA	"Food and Drug Administration today granted accelerated approval to Tudriqev (vusolimogene oderparepvec-wtpg), a genetically modified oncolytic viral therapy for the treatment of advanced, refractory melanoma. Tudriqev is indicated in combination with nivolumab for the treatment of adult patients with unresectable advanced cutaneous melanoma who experienced disease progression with a programmed death receptor-1 (PD-1)-blocking antibody-based regimen."

Page wording from the regulator's own site; read the page for the full announcement.

1 further regulator page(s) were read but are not listed: the page states no date, or a date more than 3 years old, so they cannot be shown as recent.

Safety warnings (FDA boxed warnings)

No boxed warning was found in the retrieved FDA labels of the drugs. No boxed warning on the retrieved label of: nirogacestat, toripalimab-tpzi, Ondansetron. No FDA label could be matched for: tarlatamab, talquetamab-tgvs. This is not a complete safety review: read the full label.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07578649	PF-08634404; Pembrolizumab	Phase 3	Not yet recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06991738	68Ga-DOTA-TATE PET Scan; Amino acid infusions	Phase 1/2	Not yet recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07227012	PF-08634404; Ipilimumab	Phase 1/2	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07023289	Telisotuzumab Adizutecan; Standard of Care	Phase 2	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06694454	azacytidine; carboplatin	Phase 1/2	Suspended (2026-10-08)	Registry status SUSPENDED; no effectiveness conclusion.
NCT06038578	TRK-950; Ramucirumab	Phase 2	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04968106	Doxorubicin; Ifosfamide	Phase 2	Active not recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05477615	Lazertinib; Pemetrexed	Phase 2	Terminated (2026-10-08)	Terminated early (Slow accrual; enrollment was closed after 14 of 28 planned participants.); no results retrieved.
NCT06925737	Ifinatamab deruxtecan; Docetaxel	Phase 3	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05312411	SCRI-E2CAR_EGFRtv1; UB_TT170	Phase 1	Active not recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.

2 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Kalinsky et al. (2025)**, randomized controlled trial: "At the primary analysis (258 events), the hazard ratio (HR) was 0.73 (95% CI, 0.57 to 0.95; nominal P = .017), with median PFS 6.0 (95% CI, 5.6 to 8.6) versus 5.3 (95% CI, 3.7 to 5.6) months and 6-month PFS rates of 50% and 37% in the abemaciclib + fulvestrant and placebo + fulvestrant arms, respectively." [7]
2. **Janjigian et al. (2025)**, randomized controlled trial: "Two-year event-free survival (Kaplan-Meier estimate) was 67.4% among the participants in the durvalumab group and 58.5% among those in the placebo group (hazard ratio for event or death, 0.71; 95% confidence interval [CI], 0.58 to 0.86; P<0.001)." [8]
3. **Pusztai et al. (2024)**, randomized controlled trial: "The corresponding hazard ratios (95% confidence intervals) for EFS were 0.70 (0.38-1.31), 0.92 (0.39-2.20), 0.52 (0.32-0.82), and 1.24 (0.69-2.23)." [9]
4. **Qiu et al. (2024)**, randomized controlled trial: "Of 1657 patients screened between 13 December 2018 and 9 February 2021, 660 were ineligible due to not meeting the eligibility criteria, withdrawal of consent, adverse events, or other reasons." [10]
5. **Herrera et al. (2024)**, randomized controlled trial: "Of 994 patients who underwent randomization, 970 were included in the intention-to-treat population for efficacy analyses." [11]

Conflicting evidence

None was retrieved.

What remains under investigation

- **Immunotherapy biomarkers:** Liquid biopsy strategies (e.g., peptide-nanobeads) to predict PD-1 inhibitor response in liver cancer are under investigation, but efficacy remains exploratory (2027).
- **Radiotherapy fractionation:** Optimal hypofractionated proton therapy regimens for high-risk lung cancer are being evaluated, with long-term outcomes pending (2027).
- **Exercise in geriatric oncology:** While observational data suggest benefits, randomized trials are needed to confirm efficacy in improving prognosis for older cancer patients (2026).
- **Non-invasive diagnostics:** Breath-based (VOCs) and EV spectroscopy for early cancer detection are promising but require multicenter validation (2027).
- **Complementary therapies:** Music therapy shows symptom management promise, but its role in routine cancer care remains understudied in large-scale trials.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- [12] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date. (*indirect evidence: different or related population*)
- [3] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date. (*indirect evidence: different or related population*)
- [1] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date. (*indirect evidence: different or related population*)
- [4] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date. (*indirect evidence: different or related population*)
- [13] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date. (*indirect evidence: different or related population*)
- [14] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date. (*indirect evidence: different or related population*)
- [15] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date.
- [2] carries a journal issue date of 2027 (later than today); treat its date as an advance-publication date.
- 12 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.

Sources

1. Li X, Tian S, Fan L, et al. Peptide-functionalized magnetic nanobeads for EpCAM-targeted CTC isolation and immunotherapy efficacy evaluation in liver cancer. *Talanta*. 2027 Feb 1. [PMID 42632117](#). DOI 10.1016/j.talanta.2026.130466.
2. Allen AJ, Simone CB 2nd, Sun K, et al. Hypofractionated and stereotactic body proton radiotherapy for a high-risk lung cancer cohort: A multi-institutional experience from the proton collaborative group prospective registry. *Clinical and translational radiation oncology*. 2027 Jan. [PMID 42787797](#). DOI 10.1016/j.ctro.2026.101291.

3. Wang Z, Liu T, Liu B, et al. A GC-MS-based exhaled VOC diagnostic model for distinguishing early-stage lung cancer from controls. *Talanta*. 2027 Feb 1. [PMID 42664649](#). DOI 10.1016/j.talanta.2026.130485.
4. Odrzywolski A, Sieminska I, Klebowski B, et al. Extracellular vesicle-derived FTIR spectroscopy enables sensitive detection of colorectal cancer compared to plasma. *Spectrochimica acta. Part A, Molecular and biomolecular spectroscopy*. 2027 Jan 15. [PMID 42748722](#). DOI 10.1016/j.saa.2026.128782.
5. Iturri A, Najún M, Vázquez N, et al. Music therapy interventions on symptom management in adults with advanced cancer - multisession longitudinal impact: systematic review. *BMJ supportive & palliative care*. 2026-08-28. [PMID 41922094](#). DOI 10.1136/spcare-2025-005913.
6. Fang W, Wang Y, Yang Z, et al. Exercise interventions remodel the aging microenvironment to improve prognosis in older patients with cancer. *Frontiers in immunology*. 2026-08-18. [PMID 42683260](#). DOI 10.3389/fimmu.2026.1785073.
7. Kalinsky K, Bianchini G, Hamilton E, et al. Abemaciclib Plus Fulvestrant in Advanced Breast Cancer After Progression on CDK4/6 Inhibition: Results From the Phase III postMONARCH Trial. *Journal of clinical oncology : official journal of the American Society of Clinical Oncology*. 2025 Mar 20. [PMID 39693591](#). DOI 10.1200/JCO-24-02086.
8. Janjigian YY, Al-Batran SE, Wainberg ZA, et al. Perioperative Durvalumab in Gastric and Gastroesophageal Junction Cancer. *The New England journal of medicine*. 2025 Jul 17. [PMID 40454643](#). DOI 10.1056/NEJMoa2503701.
9. Pusztai L, Denkert C, O'Shaughnessy J, et al. Event-free survival by residual cancer burden with pembrolizumab in early-stage TNBC: exploratory analysis from KEYNOTE-522. *Annals of oncology : official journal of the European Society for Medical Oncology*. 2024 May. [PMID 38369015](#). DOI 10.1016/j.annonc.2024.02.002.
10. Qiu MZ, Oh DY, Kato K, et al. Tislelizumab plus chemotherapy versus placebo plus chemotherapy as first line treatment for advanced gastric or gastro-oesophageal junction adenocarcinoma: RATIONALE-305 randomised, double blind, phase 3 trial. *BMJ (Clinical research ed.)*. 2024 May 28. [PMID 38806195](#). DOI 10.1136/bmj-2023-078876.
11. Herrera AF, LeBlanc M, Castellino SM, et al. Nivolumab+AVD in Advanced-Stage Classic Hodgkin's Lymphoma. *The New England journal of medicine*. 2024 Oct 17. [PMID 39413375](#). DOI 10.1056/NEJMoa2405888.
12. Martyschuk PA, Zhukova GV, Bel'skaya LV, et al. Control-referenced bacterial bioluminescence reveals a two-axis salivary response phenotype in benign and malignant breast disease. *Talanta*. 2027 Feb 1. [PMID 42691711](#). DOI 10.1016/j.talanta.2026.130526.
13. Sharma M, Kumar A, Arularasu MV, et al. A unified machine learning framework for oral cancer classification using Raman spectra from tissue, serum, and urine. *Spectrochimica acta. Part A, Molecular and biomolecular spectroscopy*. 2027 Jan 15. [PMID 42731152](#). DOI 10.1016/j.saa.2026.128705.
14. Latysheva A, Protoshchak V, Karpushchenko E, et al. Near-infrared spectroscopy for urine-based cancer detection: a systematic characterization of the analytical challenges of a complex biofluid matrix. *Spectrochimica acta. Part A, Molecular and biomolecular spectroscopy*. 2027 Jan 15. [PMID 42679547](#). DOI 10.1016/j.saa.2026.128663.
15. Liu R, Peng B, Qi K, et al. FTIR spectroscopy-based differentiation of seborrheic dermatitis and scalp psoriasis using clinically normal fingernails. *Spectrochimica acta. Part A, Molecular and biomolecular spectroscopy*. 2027 Jan 5. [PMID 42531968](#). DOI 10.1016/j.saa.2026.128518.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- [FDA approvals in lung cancer treatment - Lung Cancer Research Foundation](#)
- [MSK-Led Research Resulted in New FDA Approvals for Cancer Drugs in 2025 | Memorial Sloan Kettering Cancer Center](#)
- [A Comprehensive Overview of US-FDA-Approved Anticancer Drugs and Biologics in 2023: An Observational Study](#)
- [Oncology \(Cancer\)/Hematologic Malignancies Approval Notifications | FDA](#)

- [FDA Grants Accelerated Approval to a New Breast Cancer Treatment | FDA](#)
 - [FDA Approves First in Class Targeted Therapy for Metastatic Pancreatic Cancer | FDA](#)
 - [FDA Approves New Engineered Viral Immunotherapy for Patients with Treatment-Resistant Advanced Melanoma | FDA](#)
 - [FDA drug label: Ondansetron \(label effective 2026-07-29\)](#)
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Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.